Control

ME 350: Design & Manufacturing II

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Some slides adopted from Mike Umbriac, David Remy, Chinedum Okwudire, & Shorya Awtar

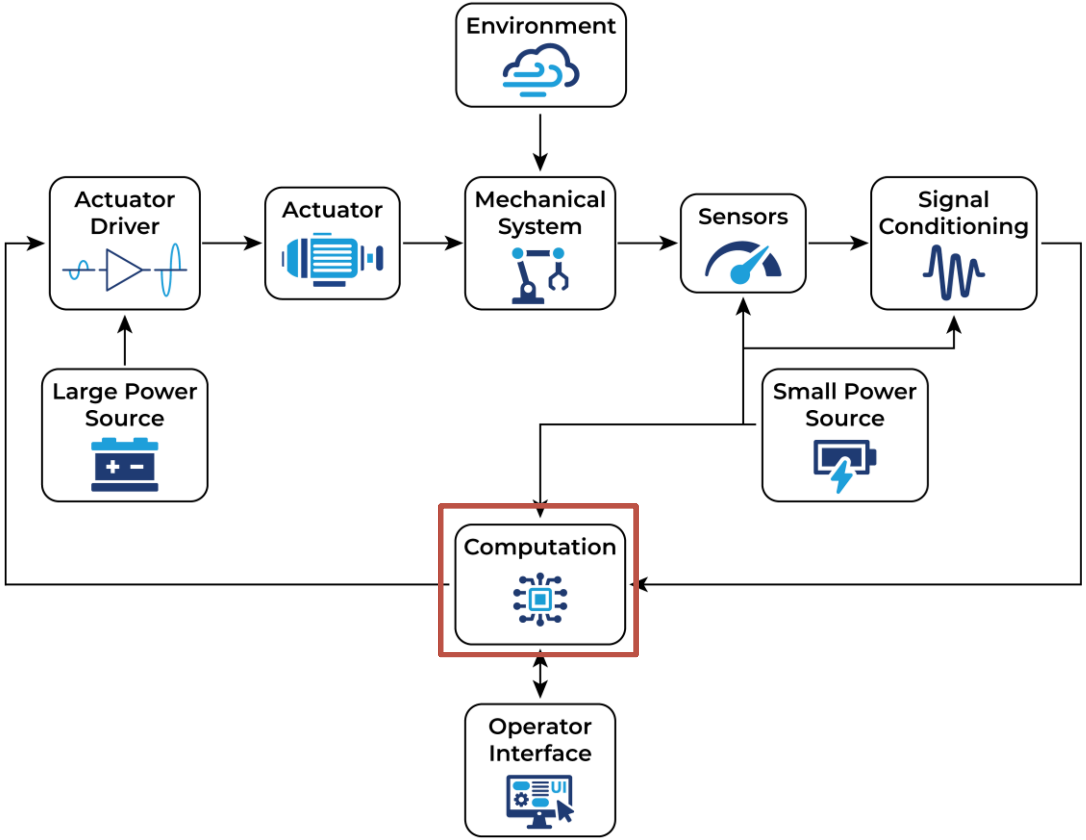
MECHANICAL ENGINEERING
UNIVERSITY OF MICHIGAN

Today's Learning Objectives

By the end of today's class you should…

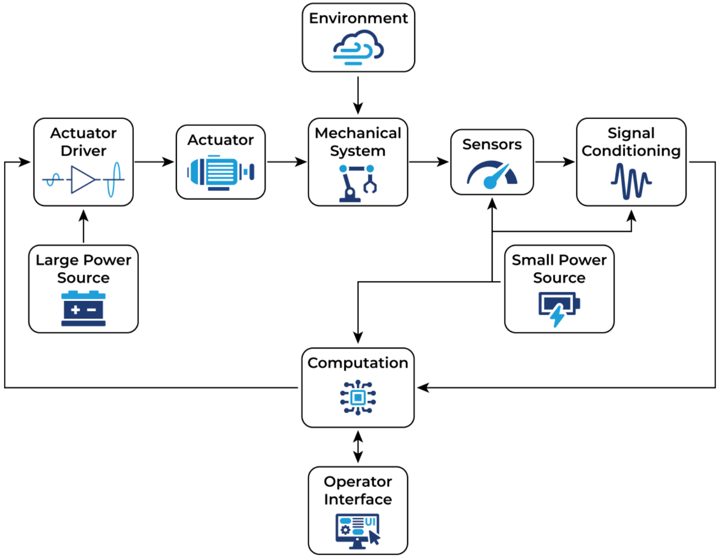
- Be able to define what "control" is from a mechatronics perspective
- Identify two major categories of control
  - Open Loop (Feedforward)
  - Closed Loop (Feedback)
- Have a high level understanding of Proportional + Integral + Derivative (PID) Control

Context for Today's Lesson



Objective of Control

- **Control:** Manipulating the **input** to a system to achieve a **desired output**
- The controller makes decisions for actuation steps which in turn determines the behavior of the system
- The decision process is called the **control logic** or **control law** or **control algorithm**
- In mechatronic systems this control logic is programmed (i.e. in C/C++) and typically executed via a digital computer (i.e. Arduino)
- Broadly, control can be described in terms of being **open-loop** or **closed-loop**



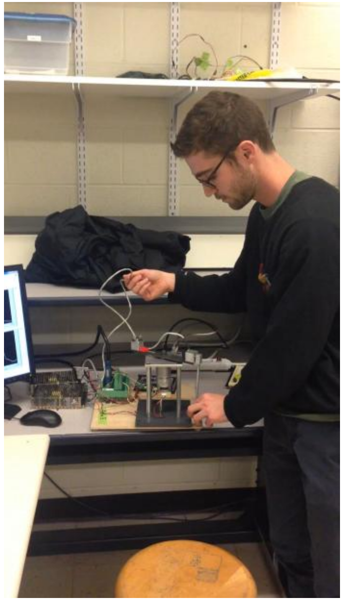
Open-loop vs. Closed-loop Control

- **Open-loop (feedforward) control:** The signal sent to our actuators is purely based upon an idea or prediction of what we think the plant will do/need


- **Closed-loop (feedback) control:** The signal to our actuators is based upon sensor measurements to achieve a desired behavior

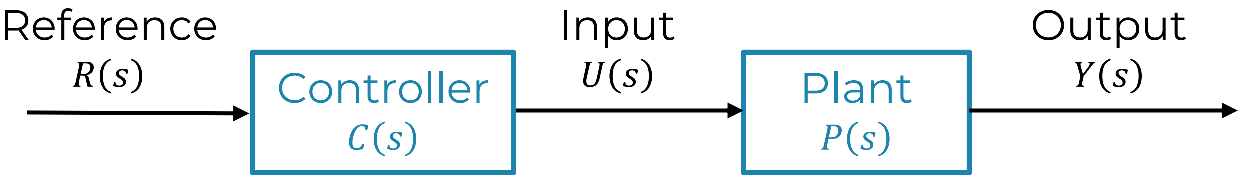
Open-loop vs. Closed-loop Control

- **Open-loop control:** The signal sent to our actuators is purely based upon an idea or prediction of what we think the plant will do/need
- **Closed-loop control:** The signal to our actuators is based upon sensor measurements to achieve a desired behavior

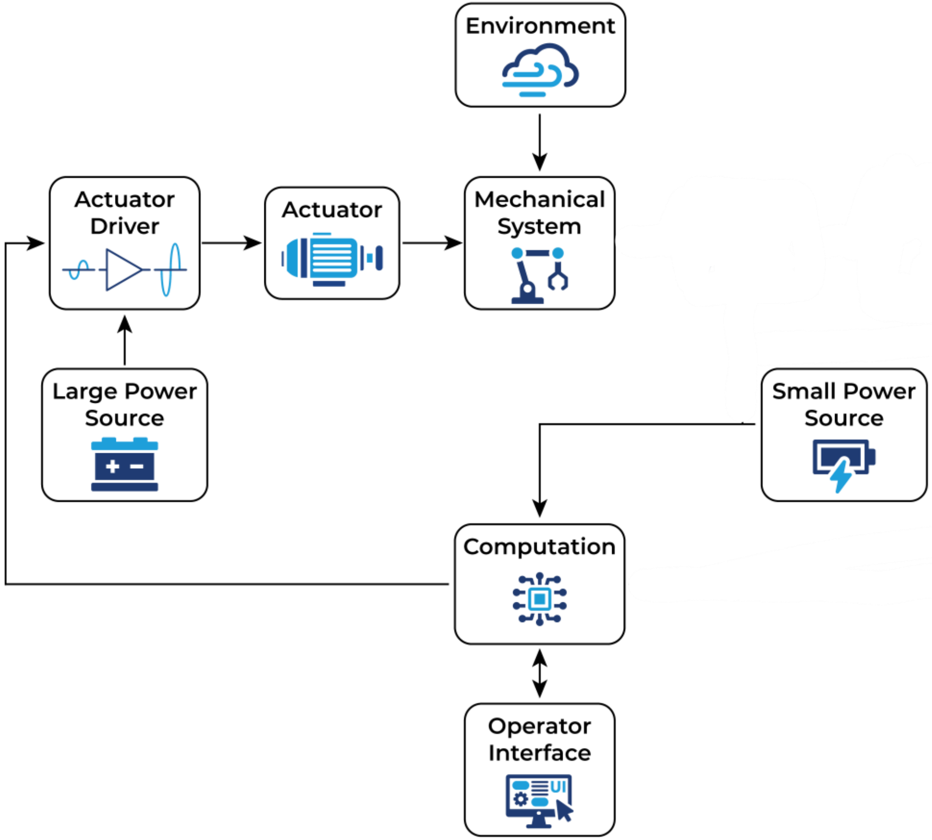


Open-loop (Feedforward) Control

- Determine a **model** of the system (plant)
  - Create a physics-based model/equations
  - Create a model via experimentation (system ID or machine learning)
- From the model, predict how the system will behave to an input
  - I.e. create a transfer function model or create a look-up table
- Select the appropriate input for a desired output using a controller
- Core idea: Choose/design a controller to be “the inverse” of the plant
  - I.e. choose $C(s)$ such that $C(s)P(s) = 1$
- **No sensor information is fed back to guide the decision**



Open-loop (Feedforward) Control



Open-loop (Feedforward) Control

MAKE THE WORLD WORK BETTER

Pros

- Simple & low cost
- No need for additional sensors or signal conditioning
- No additional noise in the system due to sensors
- No reaction time lag (preemptive or proactive)
- No stability issues*

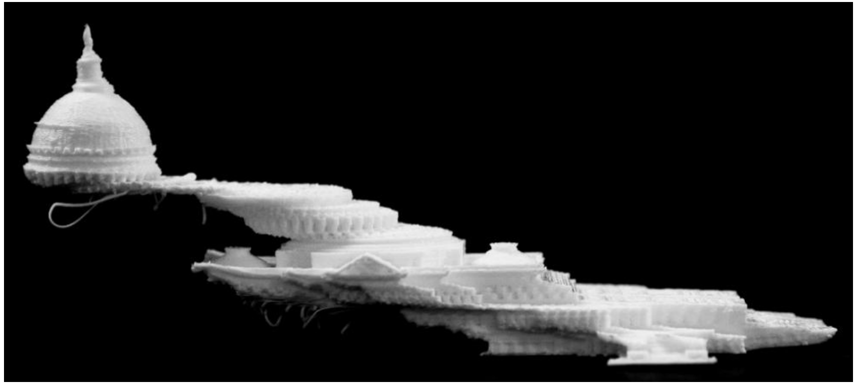
Cons

- Cannot deal with unknown situations (everything needs to be modeled & thought of)
- Cannot correct for mistakes
  - What if the model is wrong?
  - What if the system's physical properties change?
  - What if there are unanticipated disturbances?

What happens when open-loop goes wrong?

- Many desktop 3D printers don't have feedback sensing for position control and are thus open-loop relying on stepper motors to get to discrete positions
- If a stepper motor skips, there is no way to correct for it

Without skipped steps

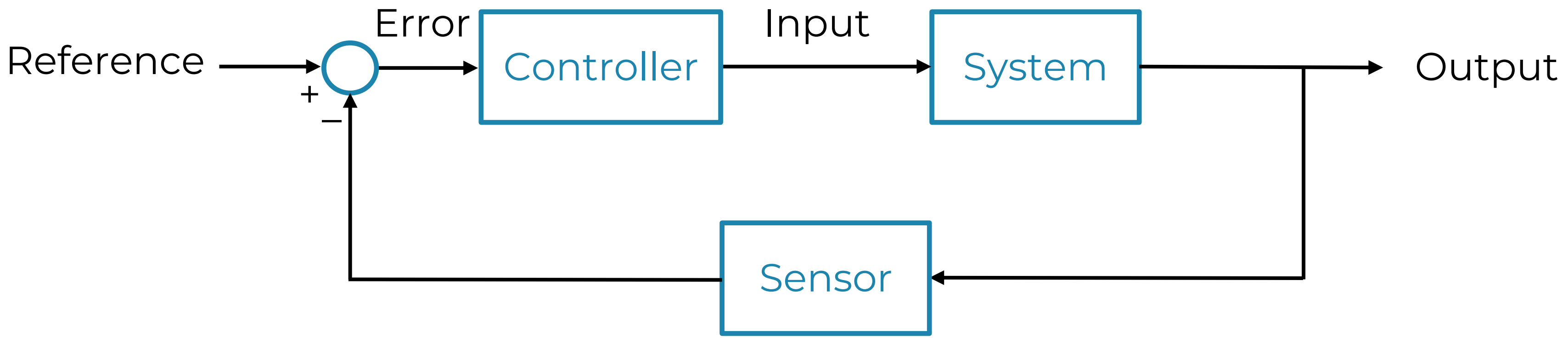


With skipped steps

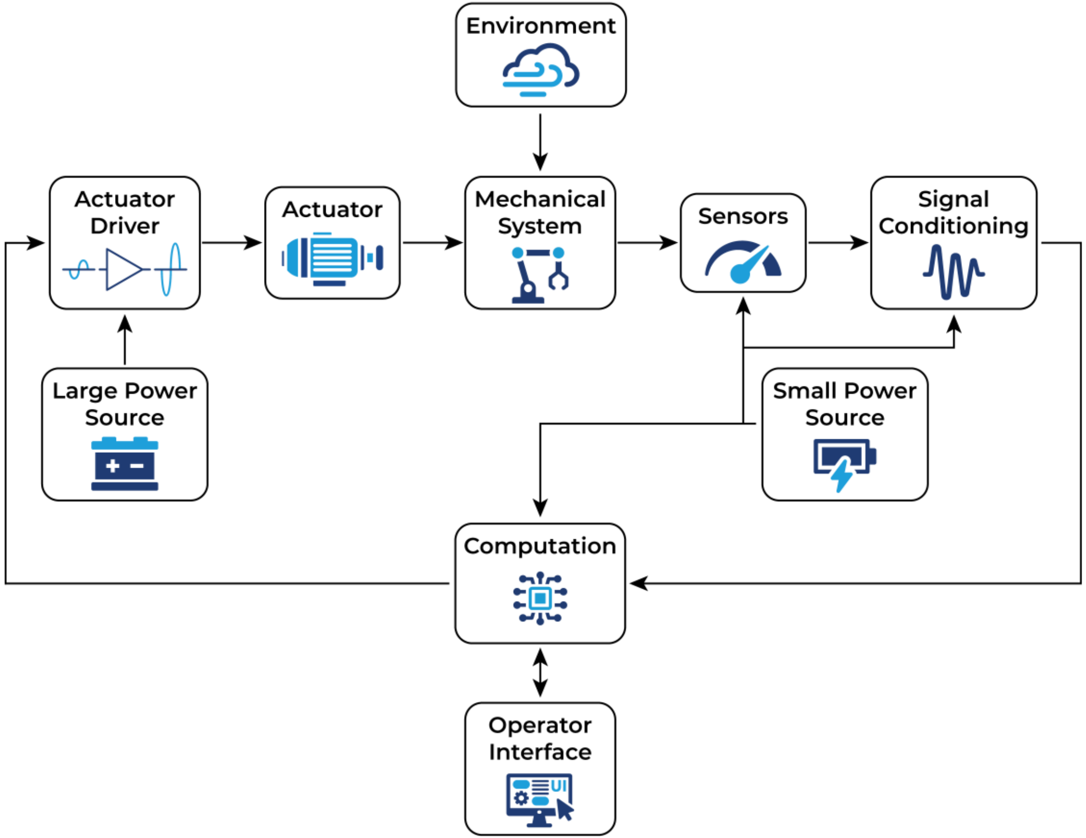
Duan, M., Yoon, D., & Okwudire, C. E. (2018). A limited-preview filtered B-spline approach to tracking control–With application to vibration-induced error compensation of a 3D printer. *Mechatronics*, *56*, 287-296.

Closed-loop (Feedback) Control

- **Measure** the quantity you wish to control using a sensor
- **Compare** with what the desired value of this quantity to get an error measurement
- **Decide** on inputs based upon this error
- A system model can always be used to tune or design appropriate feedback controllers



Closed-loop (Feedback) Control



Closed-loop (Feedback) Control

Pros

- Can deal with unknown or unplanned situations
- Can correct for mistakes

Cons

- Needs additional sensors and signal conditioning
- More complex and expensive
- Additional noise in the system due to sensors
- Is always reactive (waits for error to act)
- Can become unstable

What happens when feedback control goes right?

MAKE THE WORLD WORK BETTER

Bi-Modal Hemispherical Sensors for Dynamic Locomotion and Manipulation

Authors' Names Omitted for Anonymous Review

Demos:
Contact Following
Slip Detection
Footpad Sensor
Fingertip Sensor

What happens when feedback control goes wrong?

MAKE THE WORLD WORK BETTER

Stable Control

Unstable Control

Control in ME 350 Project

MAKE THE WORLD WORK BETTER

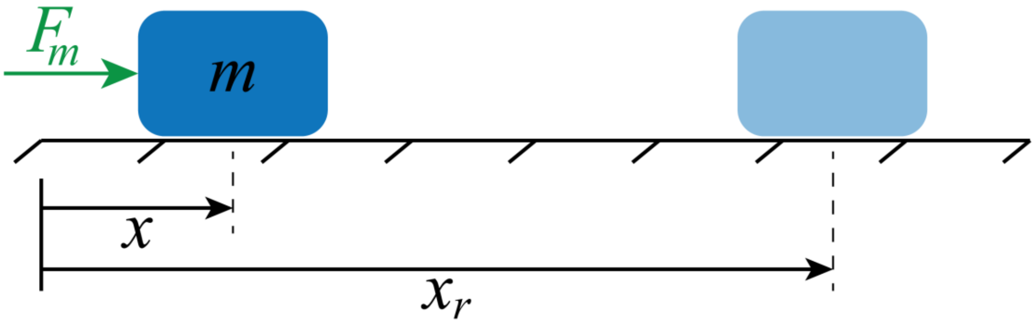
Some of the control in your project could be classified as open-loop while other aspects could be classified as closed-loop. Connect the following:

1. Using the motor encoder to determine the voltage to move motor to a target position
2. Determining the voltage to the motor to compensate friction in the linkage
3. Using proximity sensor data to determine which target position to go to
4. Calibrating your linkage system by applying a voltage to move it to a hard stop

Close-loop Control

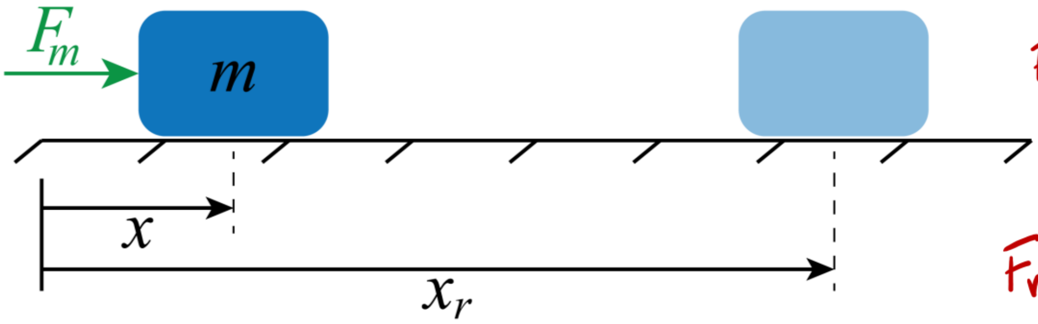
Open-loop Control

Simple Control Example – Open-Loop



We want to **design** a controller to decide the input for F_m such that the $\boldsymbol{x = x_r}$

Simple Control Example – Open-Loop

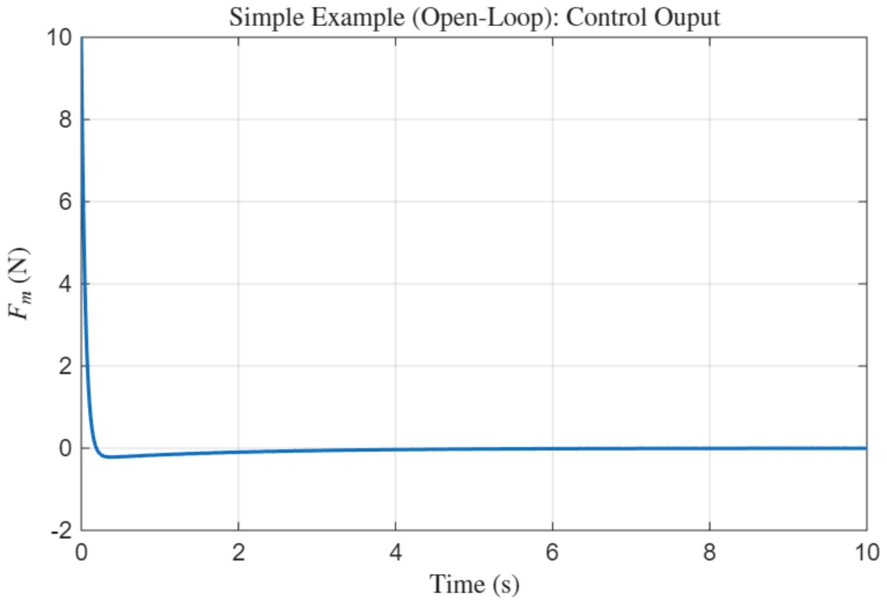


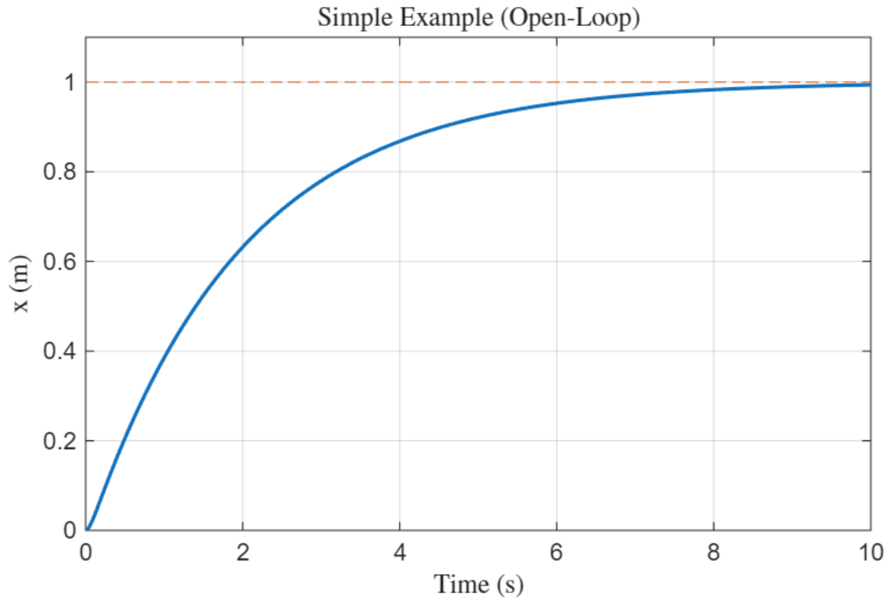
FBD

$F_m \rightarrow \square$

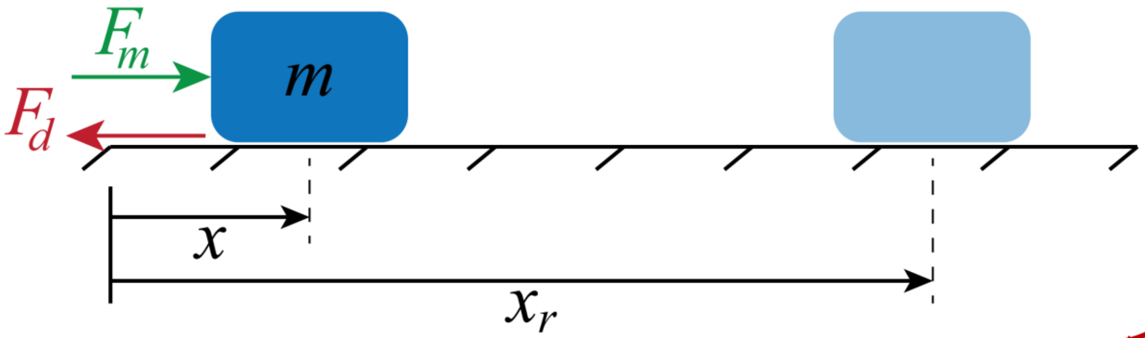
$F_m = m\ddot{x}$

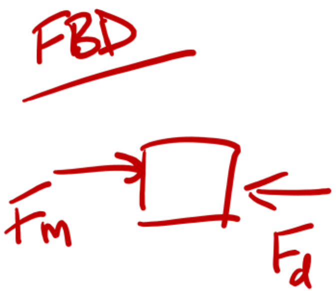
Simple Control Example – Open-Loop



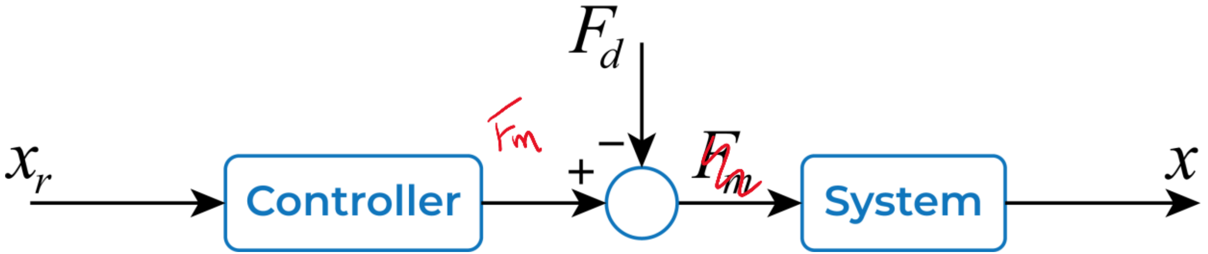


Simple Control Example – Open-Loop

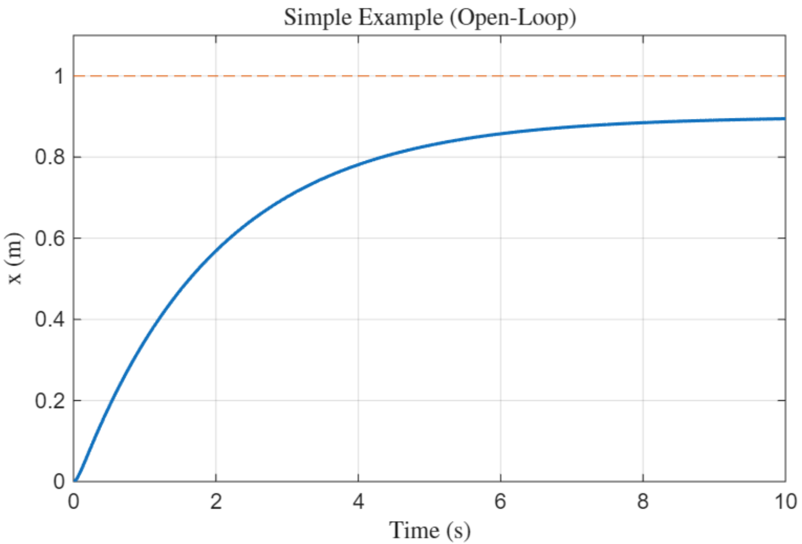




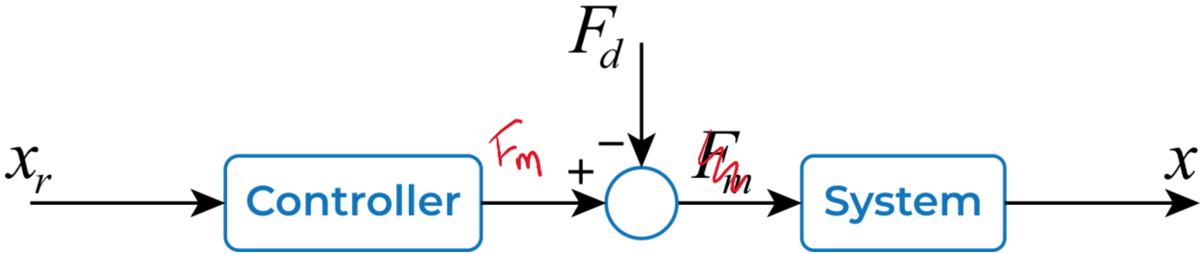
$$F_m - F_d = m\ddot{x}$$



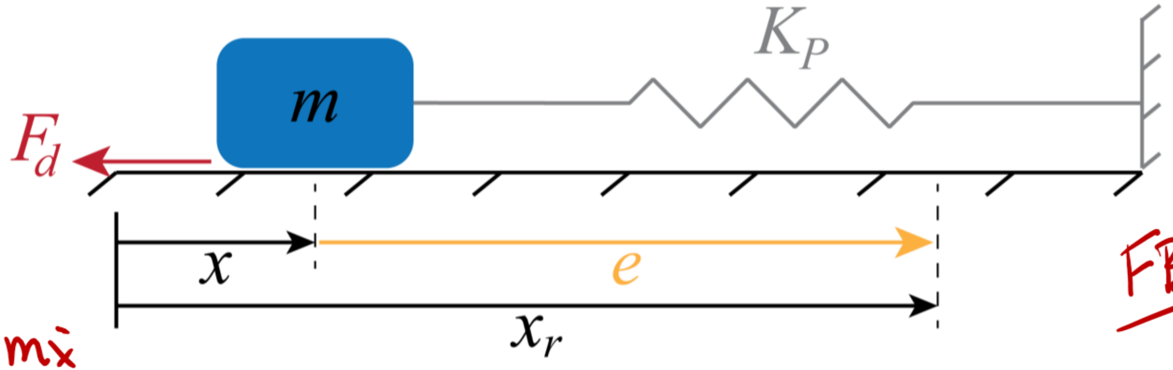
Simple Control Example – Open-Loop



We can't anticipate and design for every possible scenario!



Simple Control Example – Closed-Loop (P)



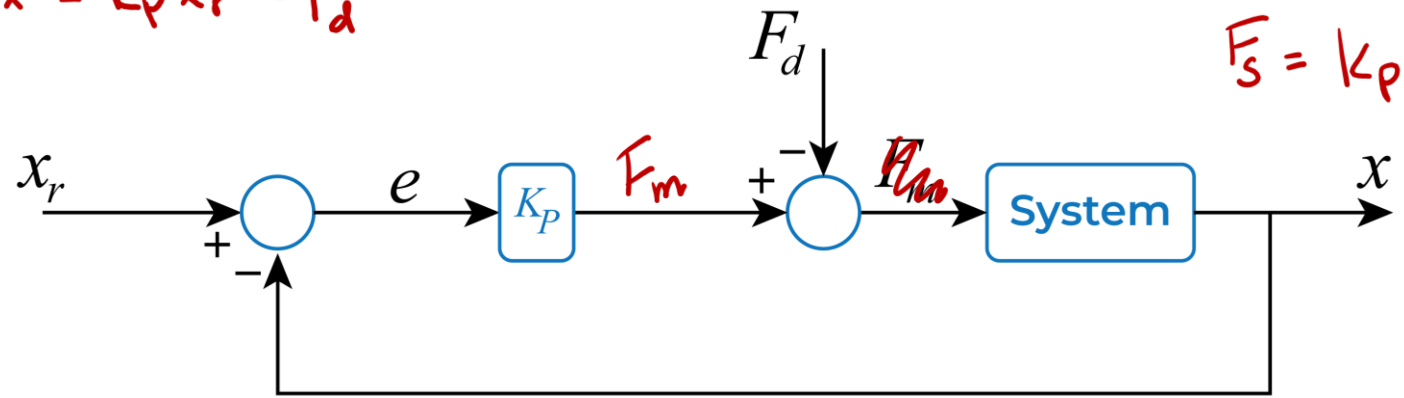
EOM

$$k_p(x_r - x) - F_d = m\ddot{x}$$

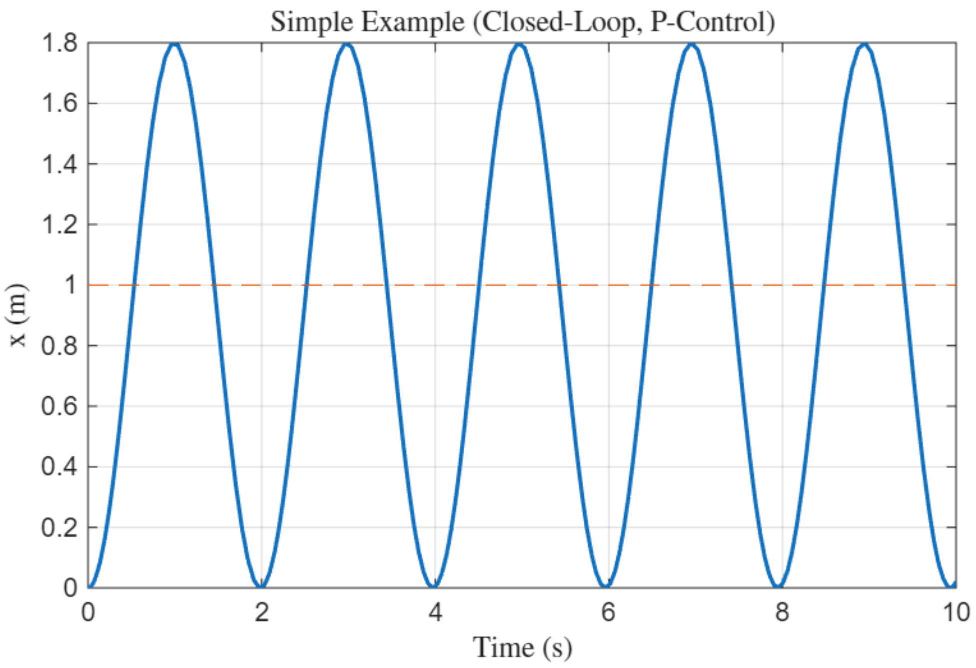
$$m\ddot{x} + k_p x = k_p x_r - F_d$$

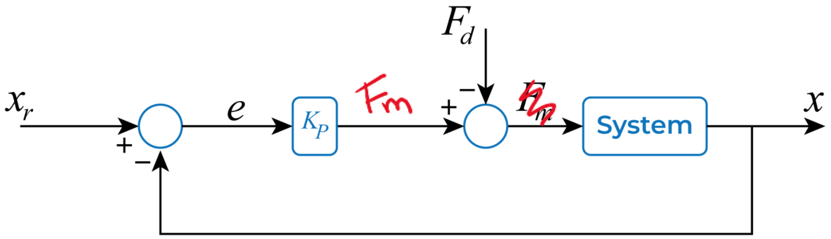
FBD

$$F_s = k_p(x_r - x)$$



Simple Control Example – Closed-Loop (P)

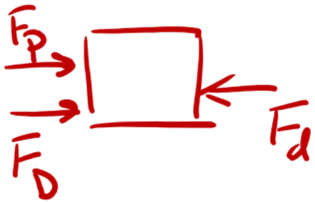


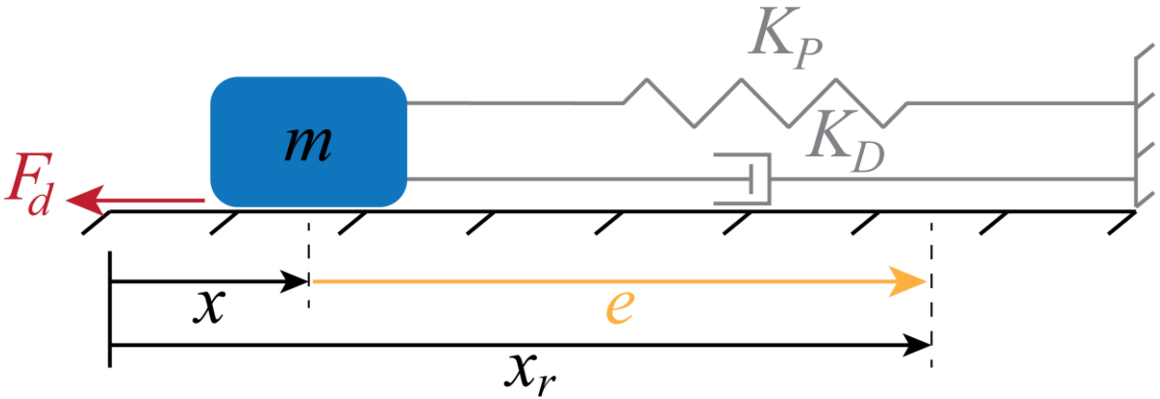


- P gain acts like a **spring**
  - High P gain increases **response speed**
- System **oscillates** about reference position (in the absence of F_d)
- If raised too high, **instability** results due to issues of delays, saturation, etc.

Simple Control Example – Closed-Loop (PD)

FBD

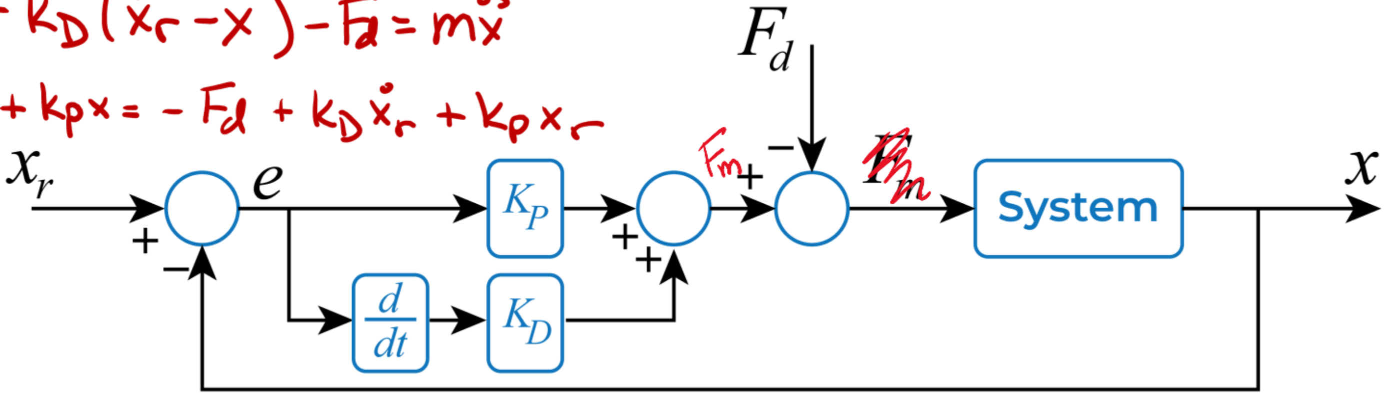




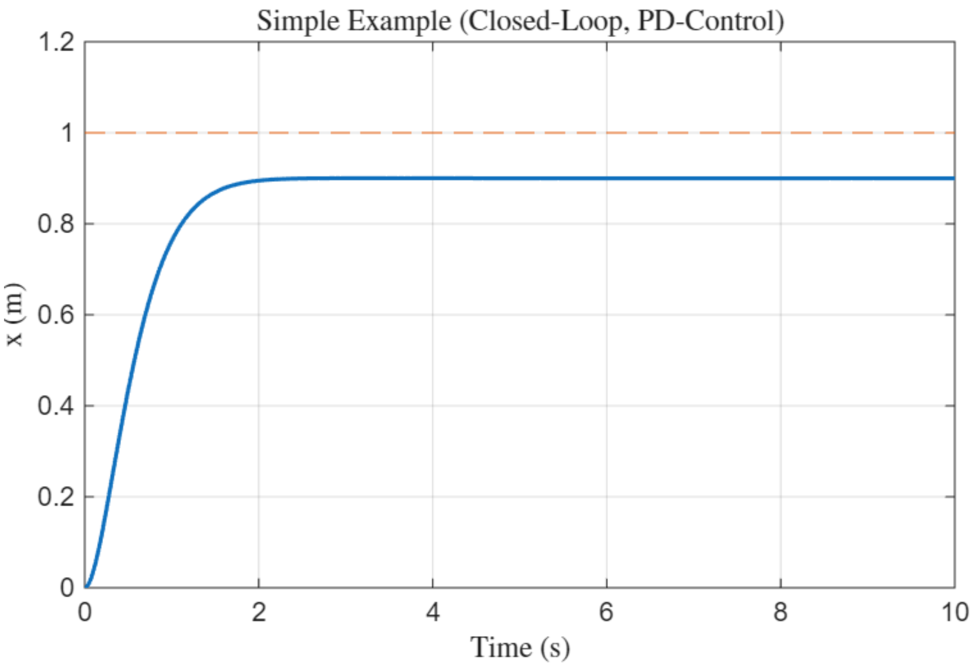
EOM

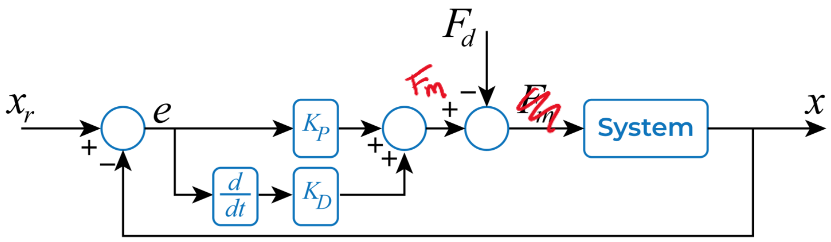
$$k_P(x_r - x) + k_D(\dot{x}_r - \dot{x}) - F_d = m\ddot{x}$$

$$m\ddot{x} + k_D\dot{x} + k_P x = -F_d + k_D\dot{x}_r + k_P x_r$$



Simple Control Example – Closed-Loop (PD)





- D gain adds **damping** to the response
- Too much D gain severely **slows down** the response and could severely amplify noise or discretization (quantization) errors

Review of Damped Systems (PD Control)

Newtons: $-kx - c\dot{x} = m\ddot{x} \Rightarrow \ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0$

ω_n is our "undamped natural freq"

General Form: $\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$

"damping ratio"

$\zeta > 1$: Overdamped

$$x(t) = A_1 e^{\lambda_1 t} + A_2 e^{\lambda_2 t}$$

$\zeta = 1$; Critically Damped

where $\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

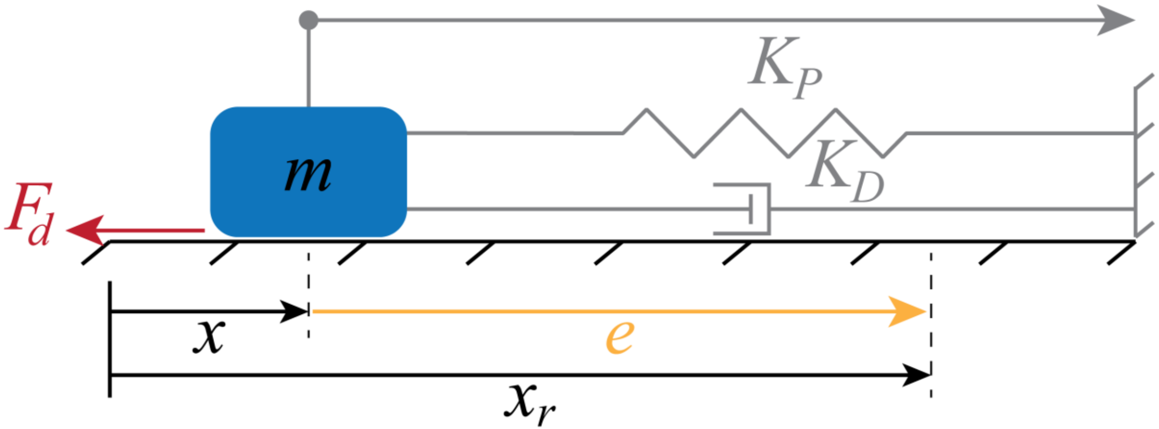
$$x(t) = A_1 e^{\lambda_1 t} + A_2 t e^{\lambda_2 t}$$

$\zeta < 1$: Underdamped System

Damped natural freq.

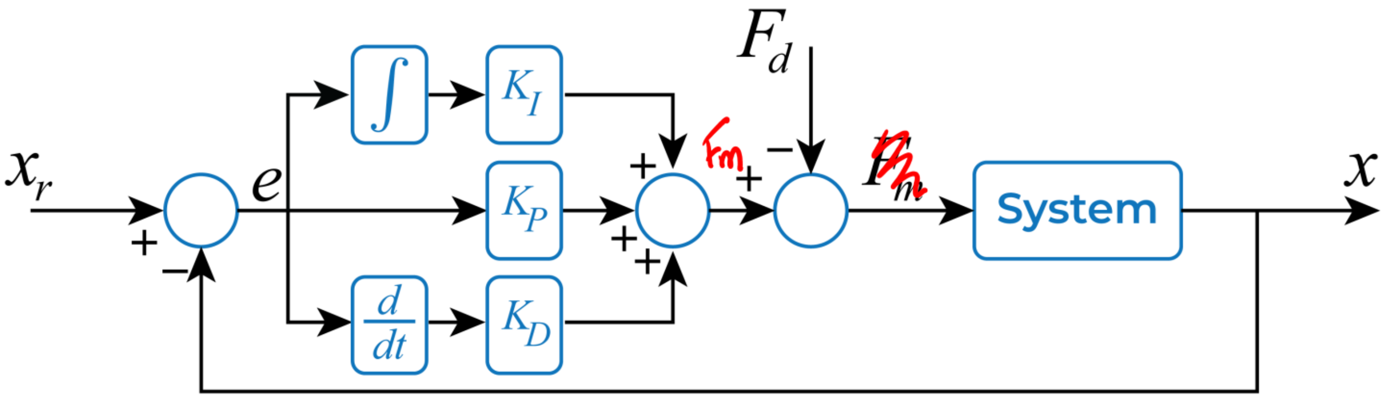
$$x(t) = e^{-\zeta\omega_n t}\left(A_1\cos(\omega_d t) + A_2\sin(\omega_d t)\right) \text{ where } \omega_d = \omega_n\sqrt{1-\zeta^2}$$

Simple Control Example – Closed-Loop (PID)

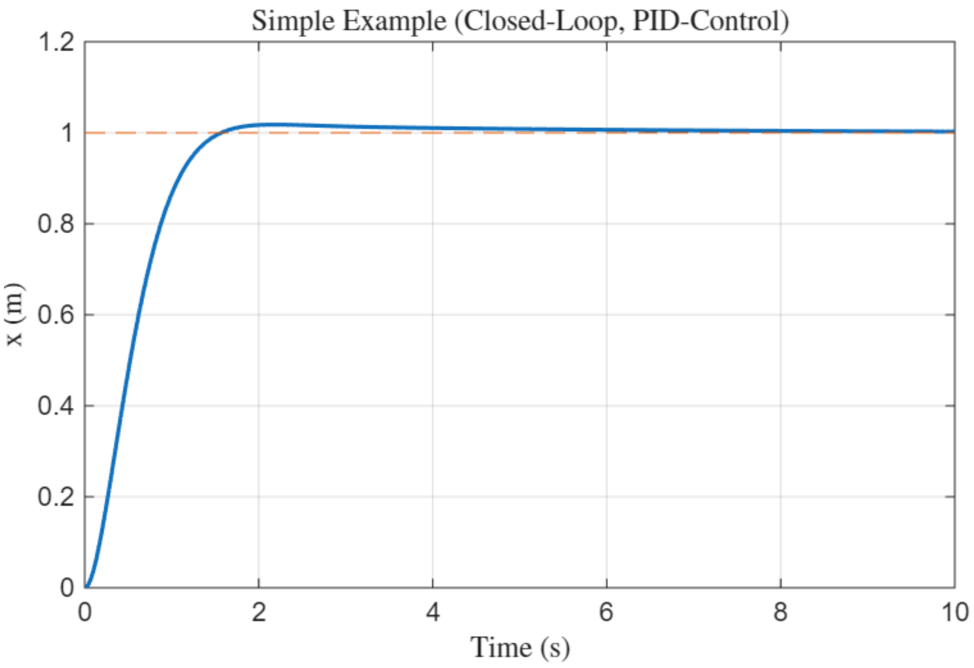


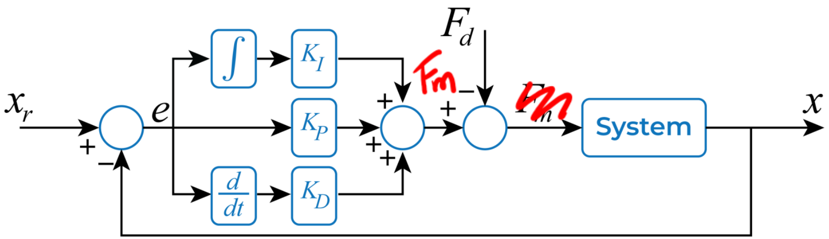
Rope is pulled until error is eliminated

Higher K_I means the rope is pulled faster and harder



Simple Control Example – Closed-Loop (PID)

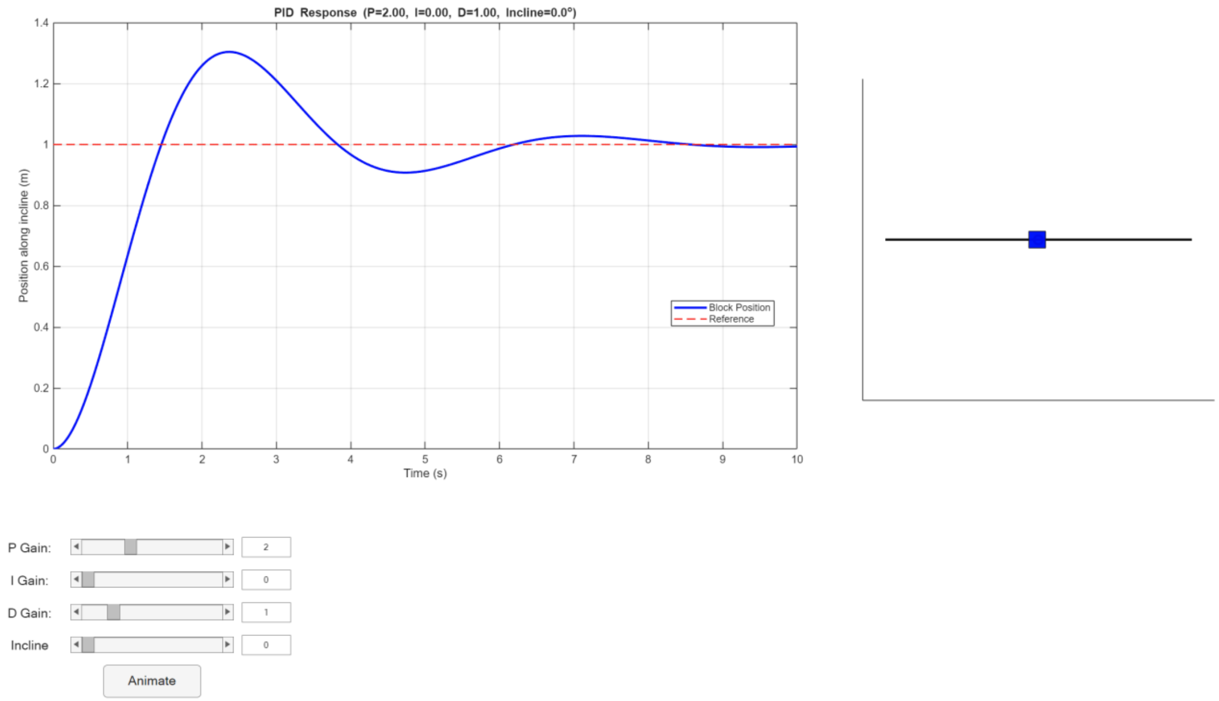




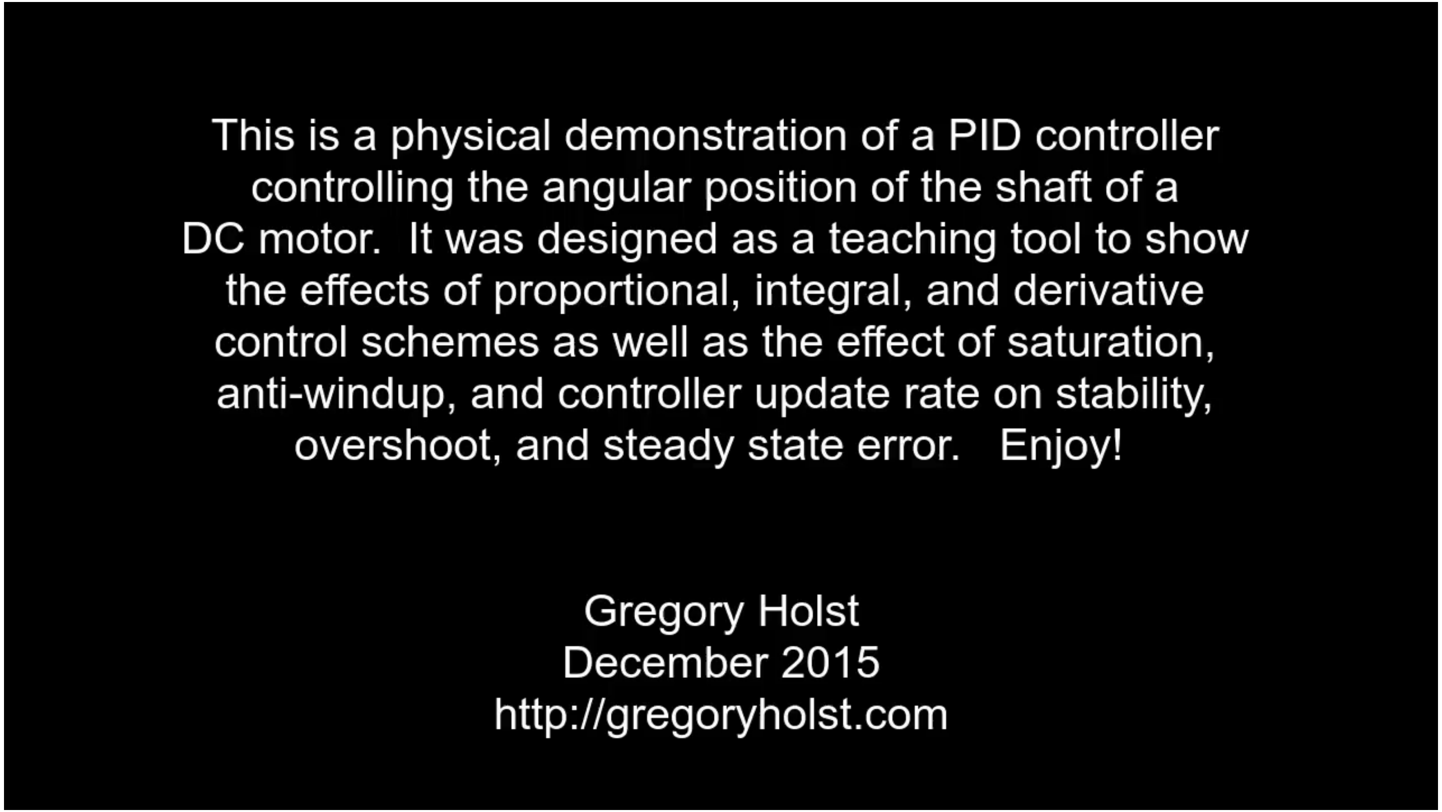
- I gain helps to eliminate **steady-state error**
- Too much I gain can cause **overshoot** and **instability**

Simple Control Example – Closed-Loop (PID)

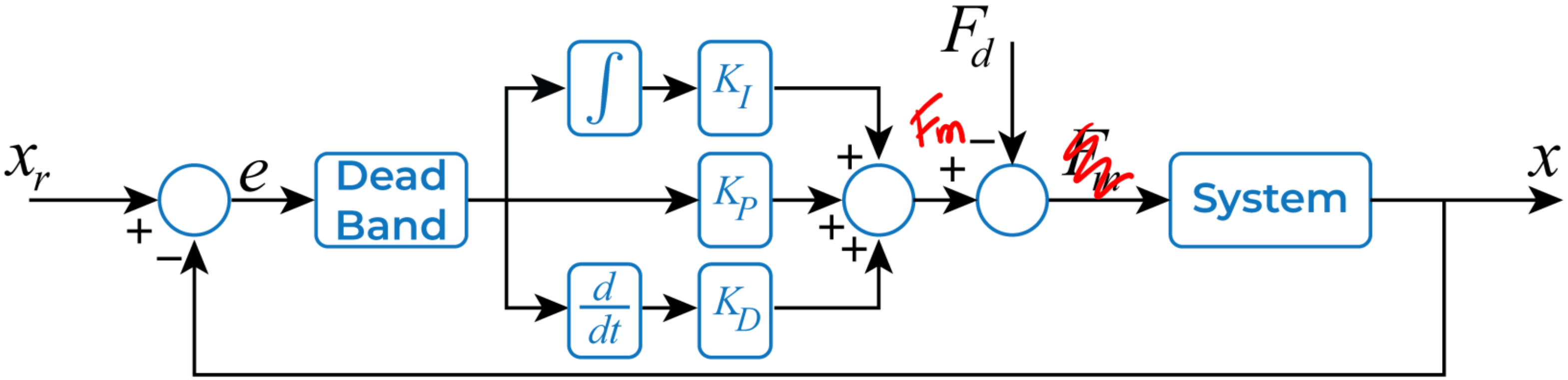
- Simple GUI to play with



PID Example in Action



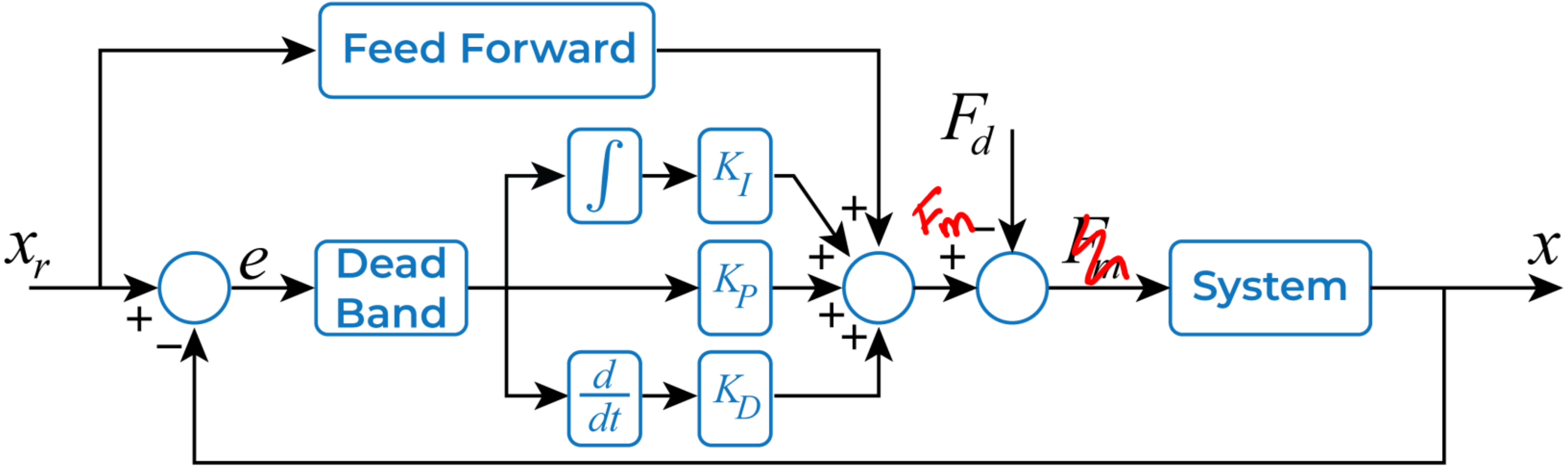
Simple Control Example – Closed-Loop (PID with Deadband)



Deadband can be introduced to reduce **chatter** (oscillations about the desired position) due to lost motion (i.e. encoder resolution, backlash)

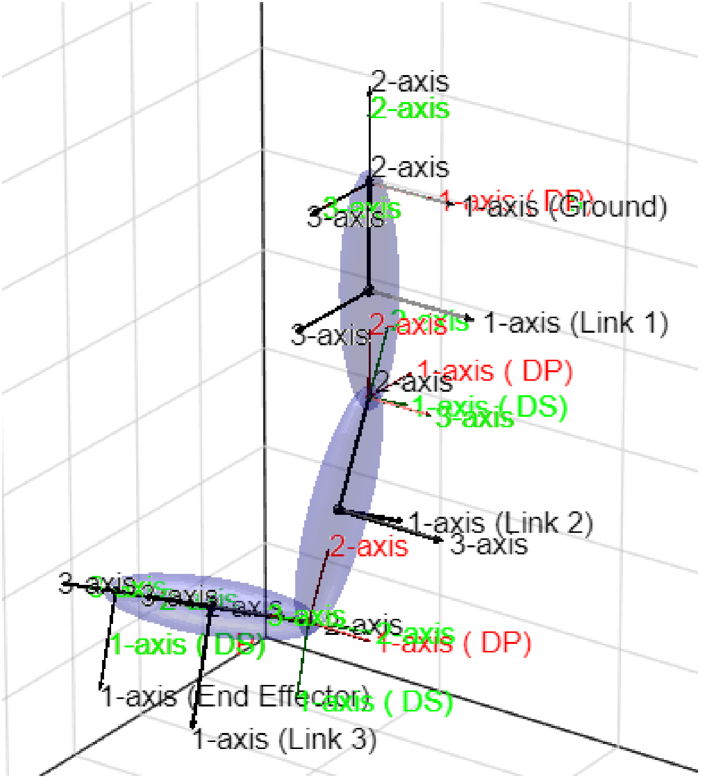
Simple Control Example – Closed-Loop (PID with Feedforward)

MAKE THE WORLD WORK BETTER

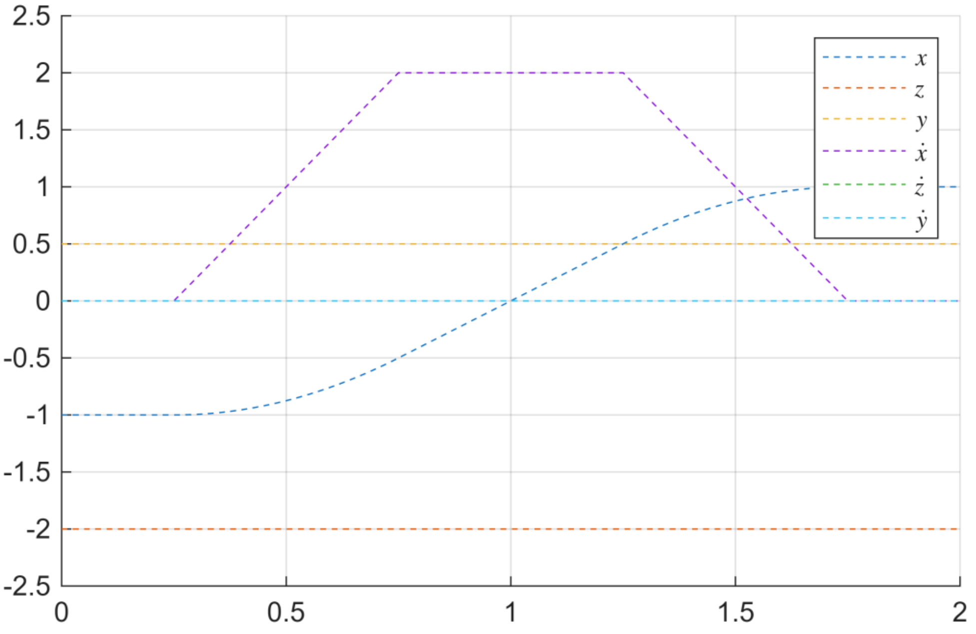


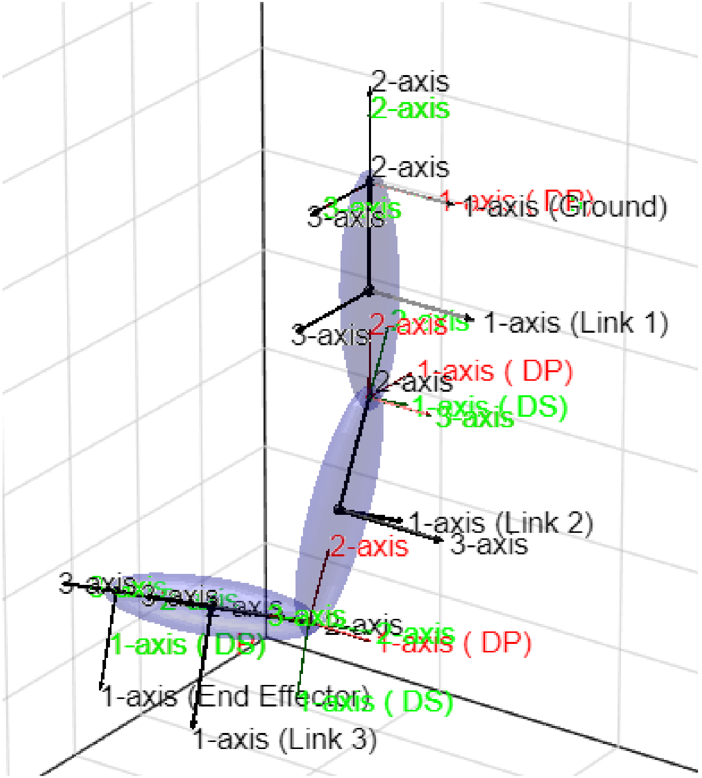
- Feedforward can make the controller **proactive** by using known information (i.e. friction/gravity) to act in advance
- Feedback helps the controller to react to **unknown** or **inaccurate** information

Control Example (ME 543)

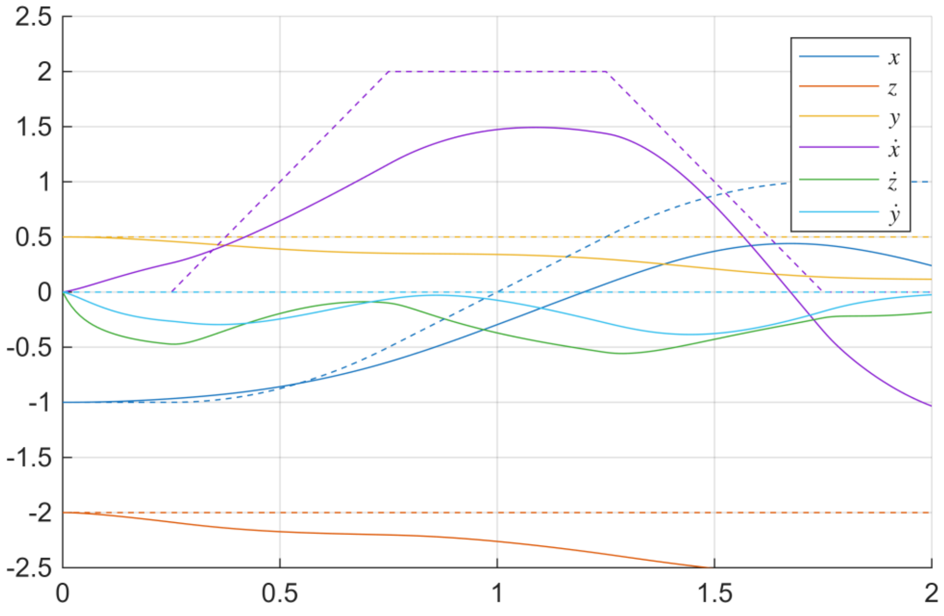


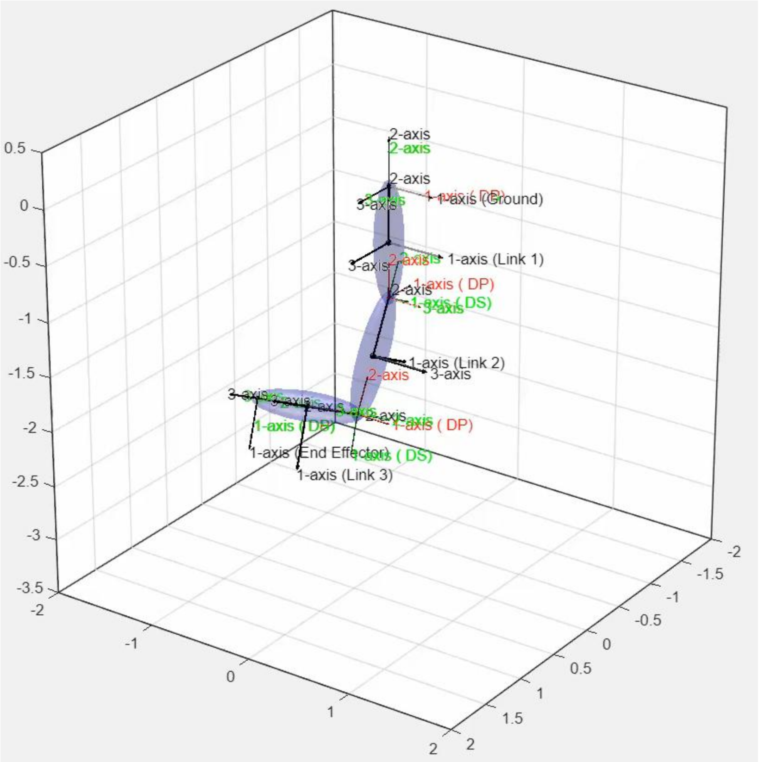
Control Example (ME 543)



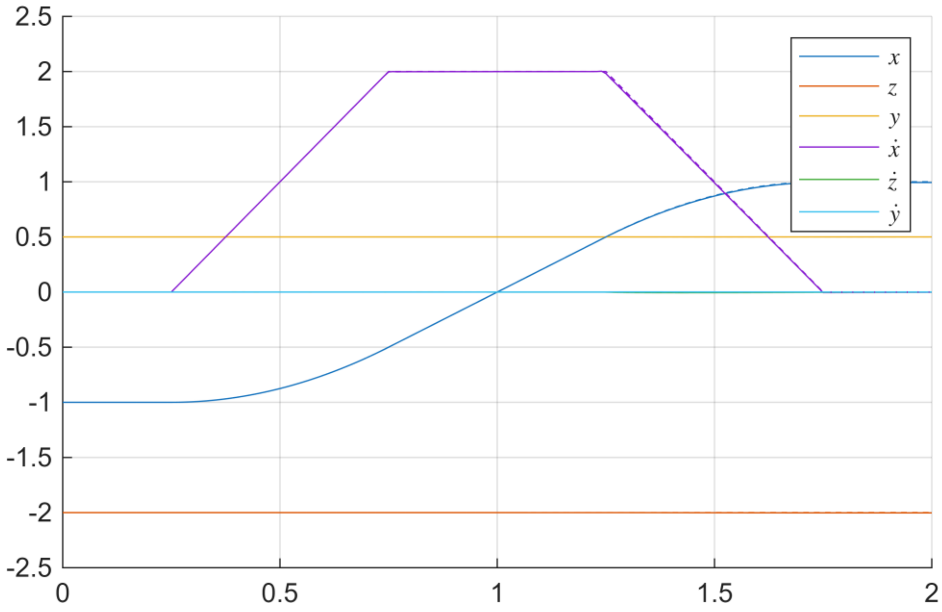


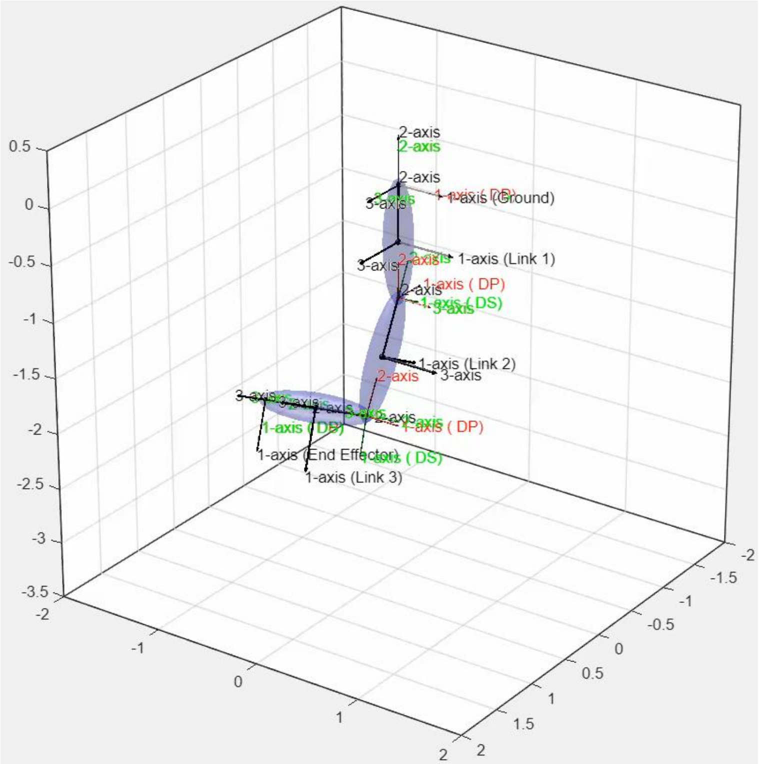
Control Example (ME 543): PD Control Only





Control Example (ME 543): PD + Feedfoward Control

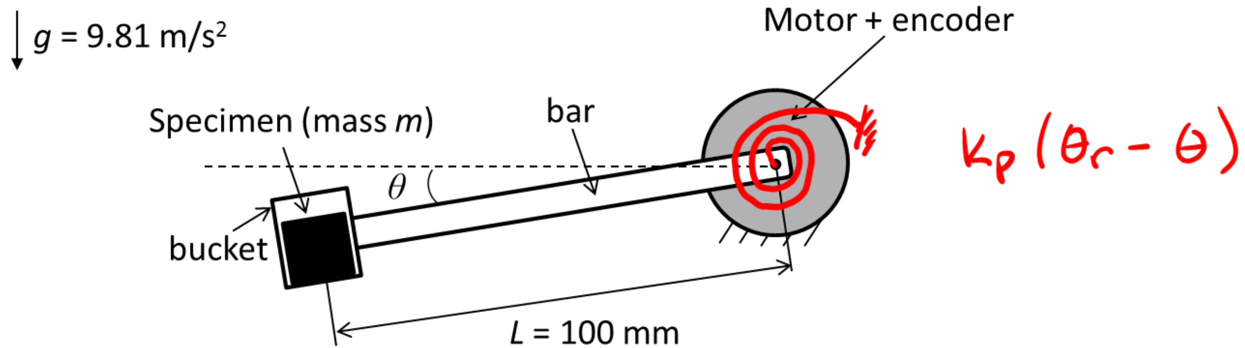




PID Tuning

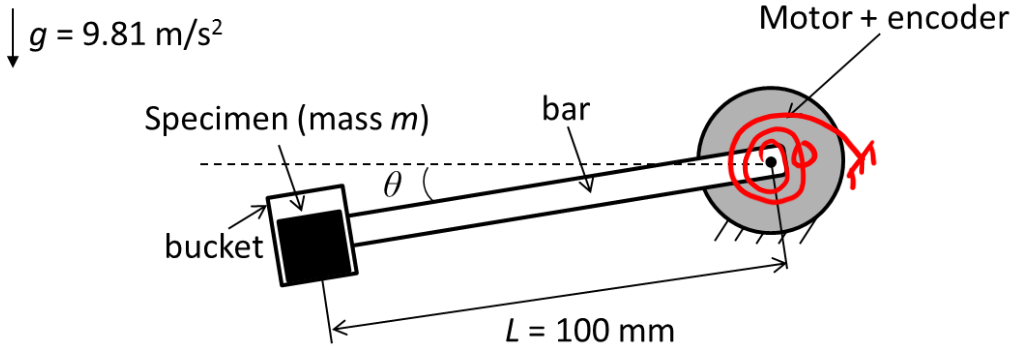
- Controller tuning is truly an **art**
- Ideally, you model your system and have a starting point from there (leaving this for ME 360...)
- This is often just a **starting point**! Why?
- Heuristic trial an error is inevitable
- Algorithmic approach
  - Ziegler-Nichols
  - Cohen-Coon

Example: P-Control of an Automatic Weighting Scale



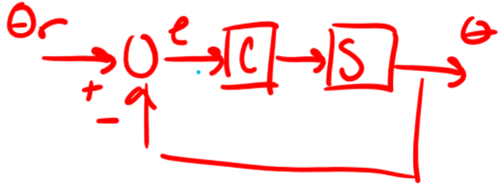
- Motor is P-controlled using encoder feedback. Encoder is zeroed before specimen is placed in bucket. Encoder reading is taken after specimen is placed in bucked and converted to mass in kg.
- We want maximum mass ($m = 10$ kg) to cause 5 degrees of rotation at equilibrium (note: 5 degrees is considered a “small rotation”)
  a) Determine the P gain off the controller in N-m/deg
  b) If encoder has a resolution of 3600 counts/rev (already quadrature), determine precision of scale in terms of the smallest mass in kg it can detect per count

Example: P-Control of an Automatic Weighting Scale



a) K_p?

$$\sum \vec{M}_o = I_o \cancel{\ddot{\theta}}^{\,0}$$



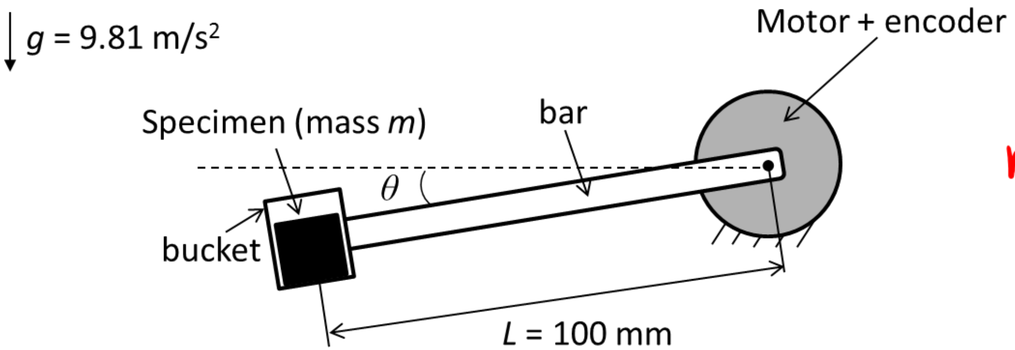
$$k_p(\cancel{\theta_r}^{\,0} - \theta)$$

$$mgL\cos\theta + T_m = 0$$

$$mgL\cos\theta - k_p\theta = 0$$

since small rotation $\cos\theta \approx 1$

Example: P-Control of an Automatic Weighting Scale



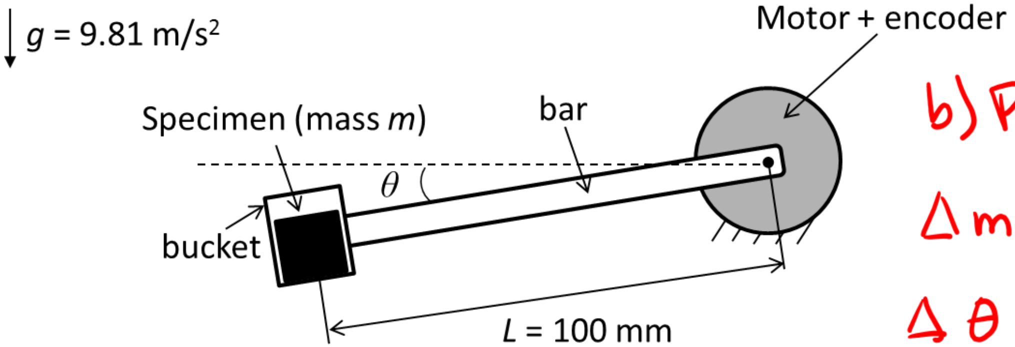
$$mgL \cancel{\cos\theta}^{\,1} = k_p \theta$$

$$mgL = k_p \theta$$

$$k_p = \frac{mgL}{\theta} = \frac{10(9.81)(0.1)}{5}$$

$$\therefore k_p = 1.962 \ \mathrm{Nm/deg}$$

Example: P-Control of an Automatic Weighting Scale



b) Precision of smallest kg per count?

Δm: mass resolution

$\Delta \theta$: position resolution

$$\Delta m g L = k_p \Delta\theta \rightarrow \Delta m = \frac{k_p \Delta\theta}{gL}$$

$$\Delta\theta = \frac{360 \text{ deg/rev}}{3600 \text{ counts/rev}} = 0.1 \text{ deg/count}$$

$$\Delta m = \frac{(1.962 \text{ Nm/deg})(0.1 \text{ deg/count})}{9.81 \text{ m/s}^2 (0.1 \text{ m})} = 0.2 \text{ kg/count}$$